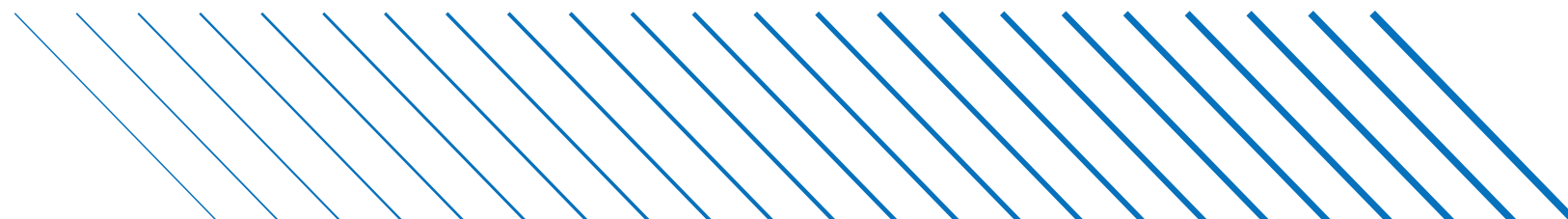
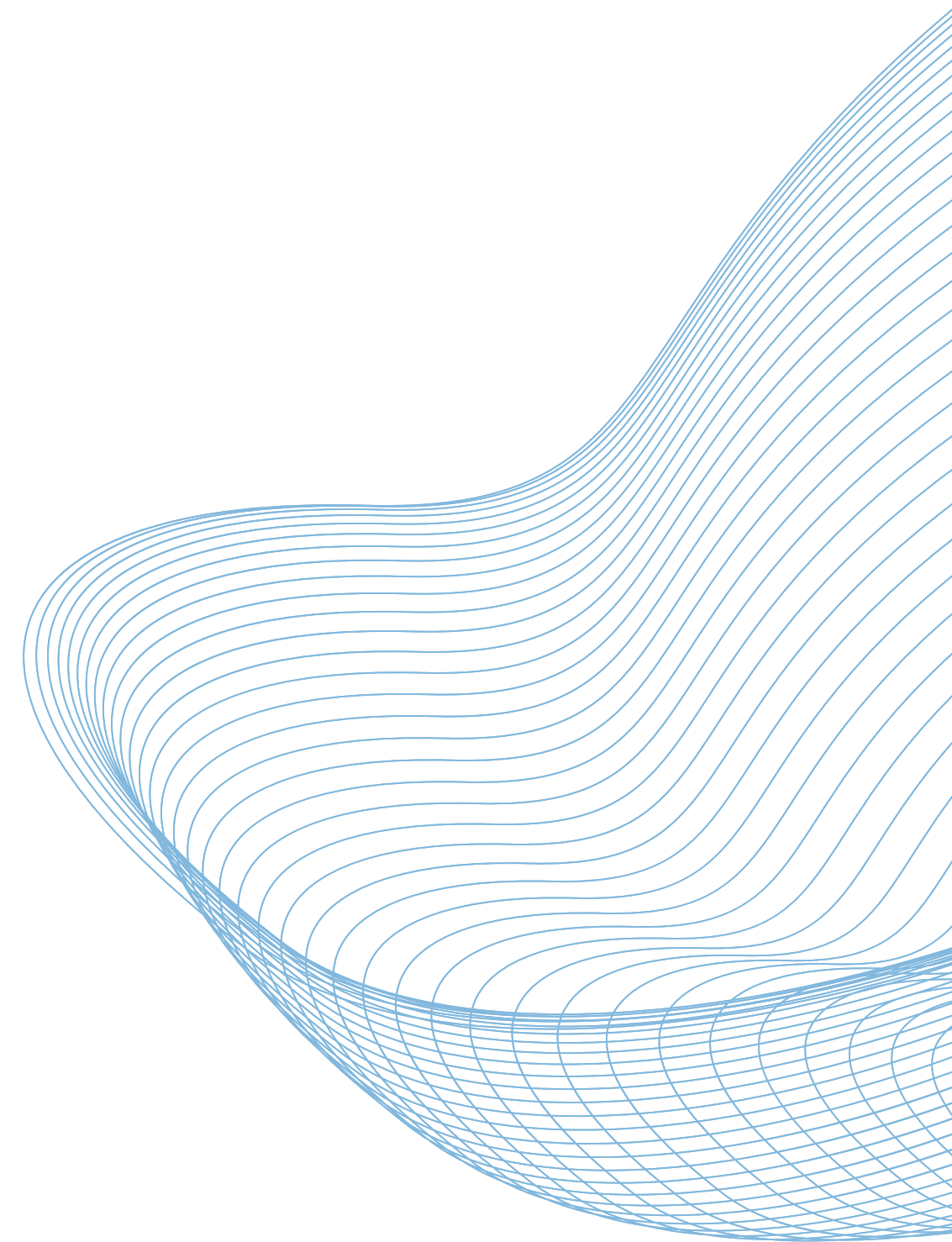


SageArk

Quant Training Session 39

Options Trading Strategy



Topics

Topic 1: Risk-Taking Using Options

Topic 2: Put-Call Parity

Topic 3: Greeks Review

Topic 4: Independent Exercise

Topic 5: Common Option Spreads

Topic 6: Types of Option Risks

Topic 7: Greeks Derivation from Option
Spreads

Topic 8: PnL of Delta-Hedged Option
Spreads

Topic 9: Independent Exercise

What to Expect from Today's Session

Topic 1: Risk-Taking Using Options

Why Trade Options?

- **Risk:** Not everyone has the appetite or personality to take risk in a price direction. Options offer non-linear risk characteristics and other risk-reward profiles.
- **Profit:** You can profit even when you have wrong spot view or when spot is range bound.
- **Trade:** You can trade the implied volatility and the risk premium priced in the option market.
- **Yield Enhancement Strategy:** It offers yield enhancement strategy to generate non-riskless passive income.

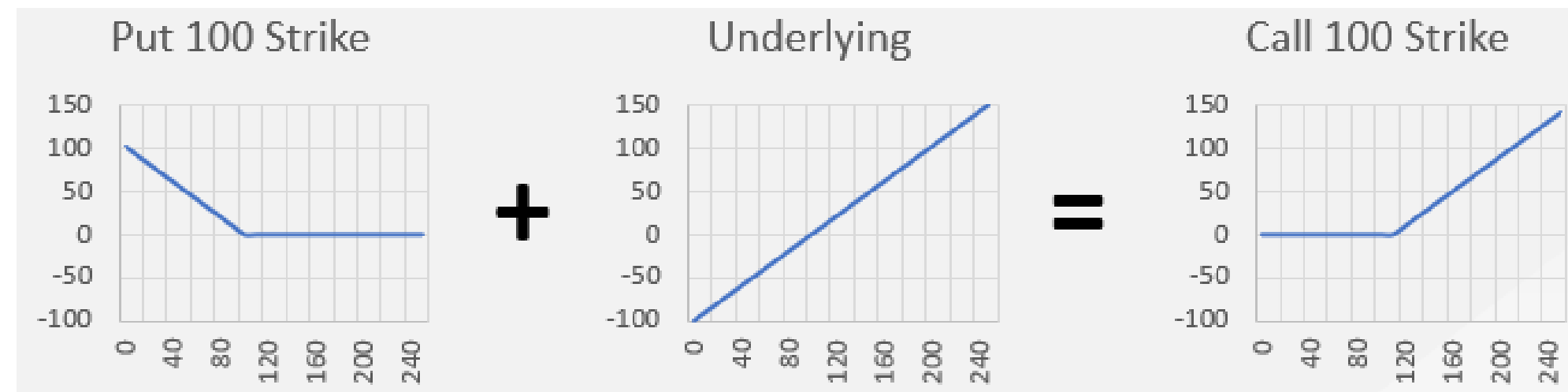


And, there are more reasons...

Topic 2: Put-Call Parity

Put-Call Parity

- **Payoff:** Payoff of long-a-call can be synthetically created by long a put (with same strike) and underlying



- **Put-Call Parity:** $\text{Call} - \text{Put} = \text{Underlying} - \text{Strike}$
- **Market Maker:** As a market maker, “most of the time,” we only care which **strikes** we own as long as they are delta-hedged
- **Call and Put:** Call and Put of the same **strike** have the same Greeks, except Delta
- **Call Strikes:** Call strikes are options with **strikes** above underlying
- **Put Strikes:** Put strikes are options with **strikes** below underlying

Put-Call Parity Example

Situation

Gold is trading at \$1800.

Question

If 1900 call is trading at \$10, where should 1900 put be priced at?

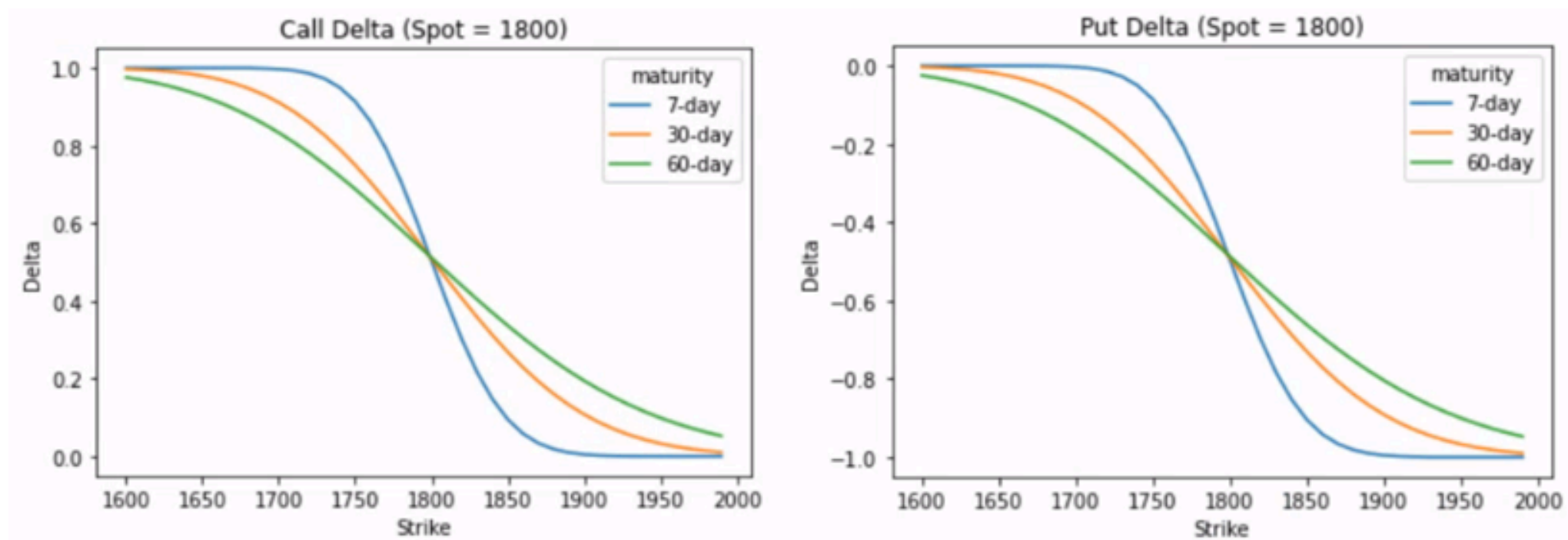
Bonus Question

What if the interest rate goes from 0% to 20%, should the price of 1900 put be higher or lower?

Topic 3: Greeks Review

Delta

- Measures option price's **sensitivity** to underlying price
- **Call** has positive delta, **Put** has negative delta

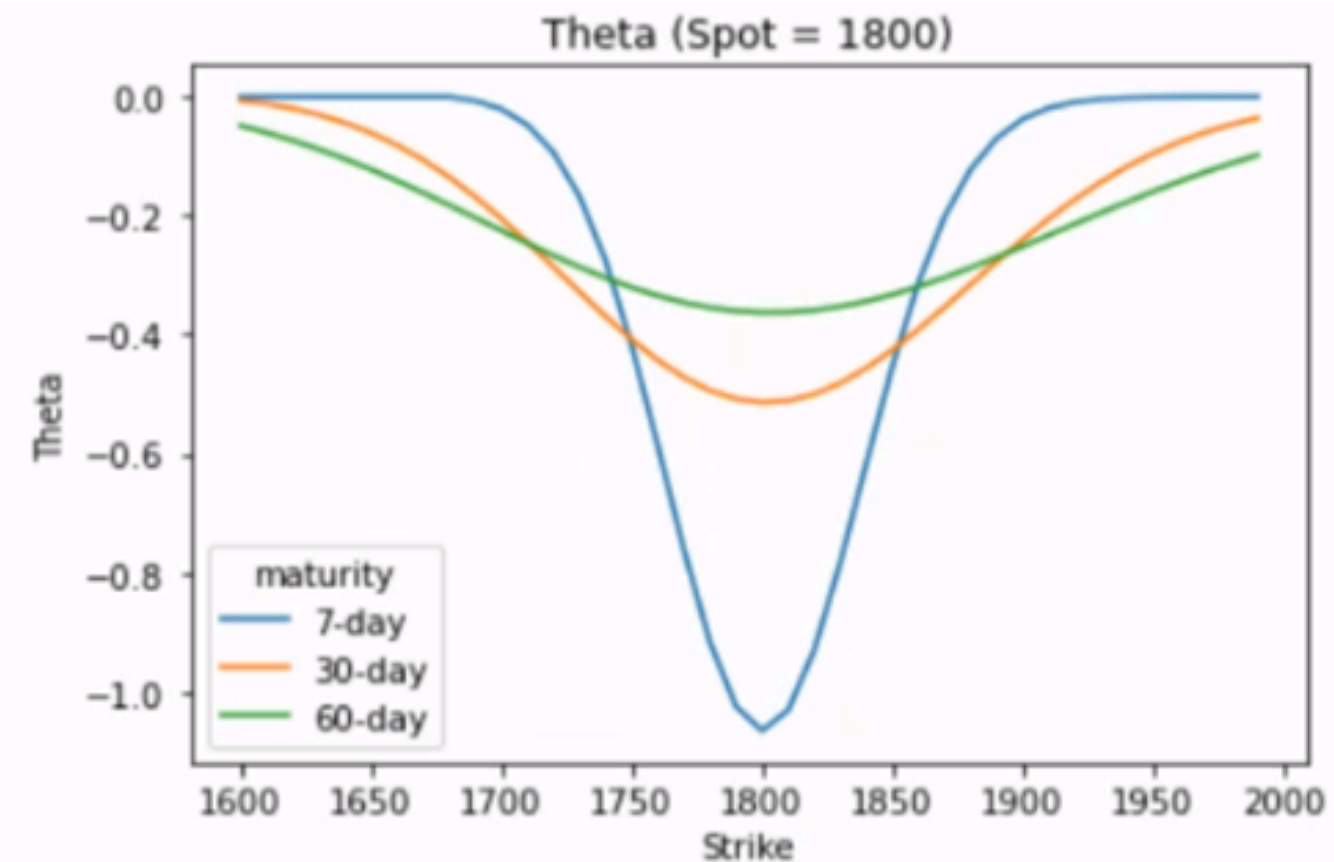
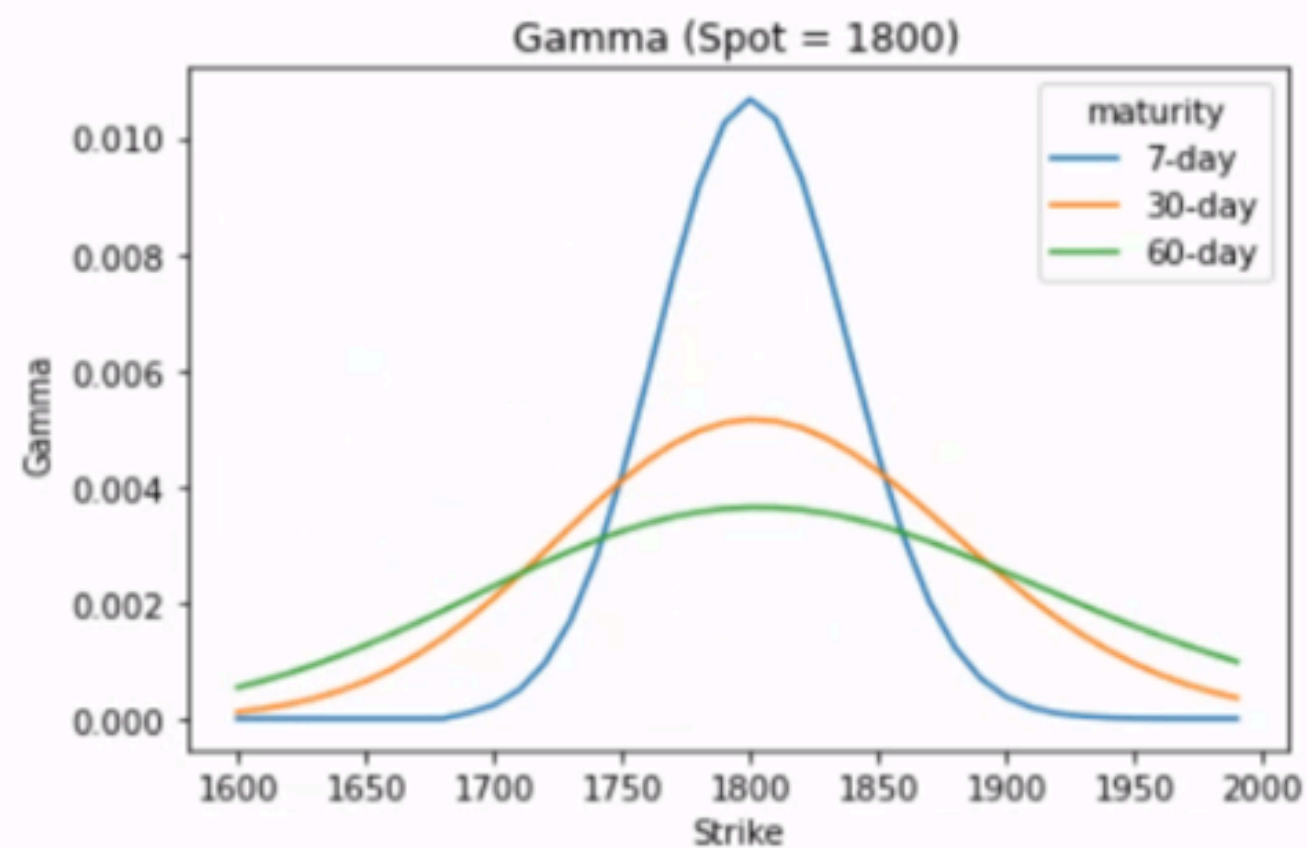


Bonus Question

Why does the delta of the call and the Abs (delta) of the put with the same strike, add up to roughly 1.0?

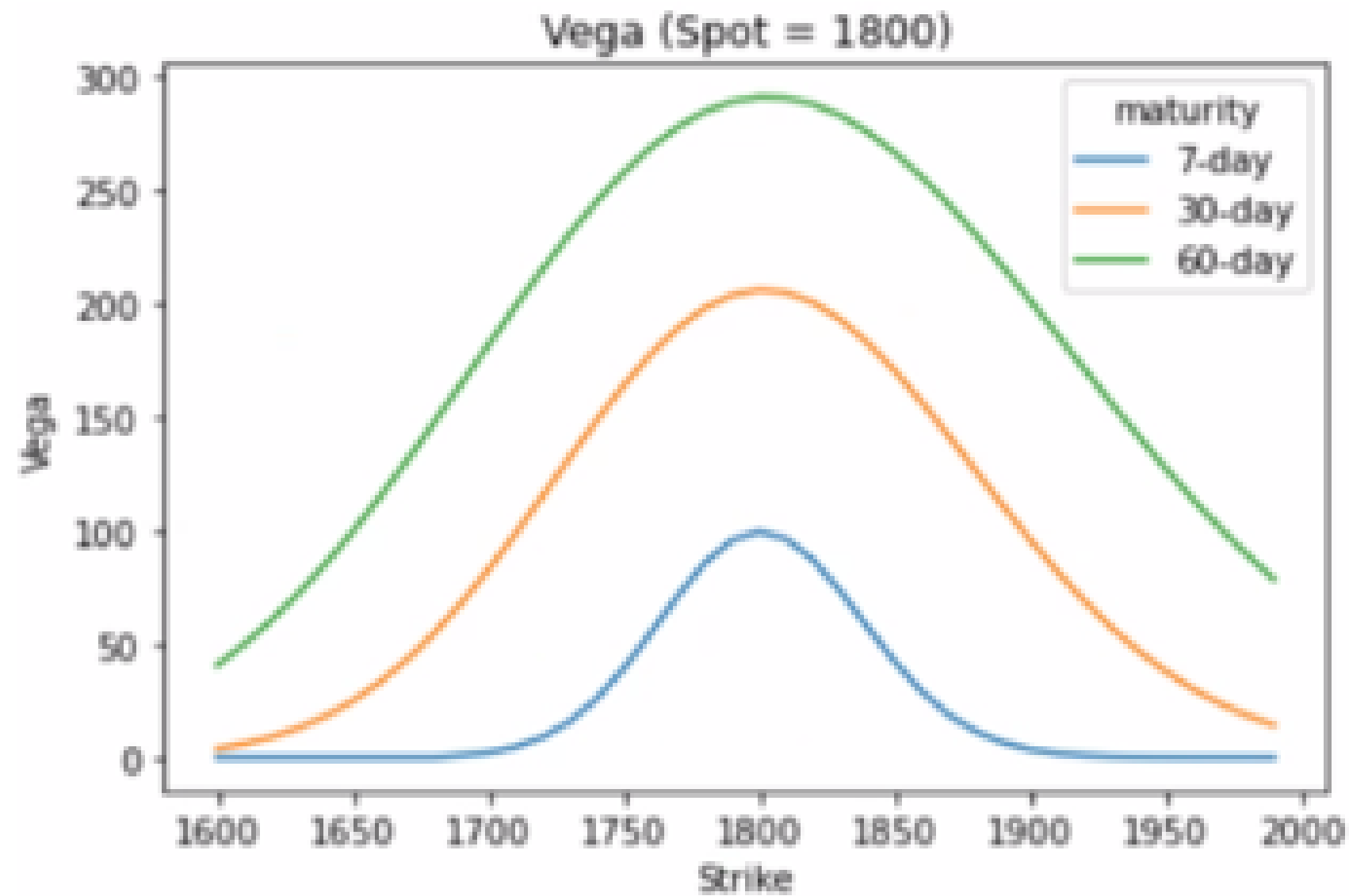
Gamma and Theta

- **Gamma:** Measures option delta's sensitivity to underlying price
- **Theta:** Measures option price's sensitivity to time



Vega

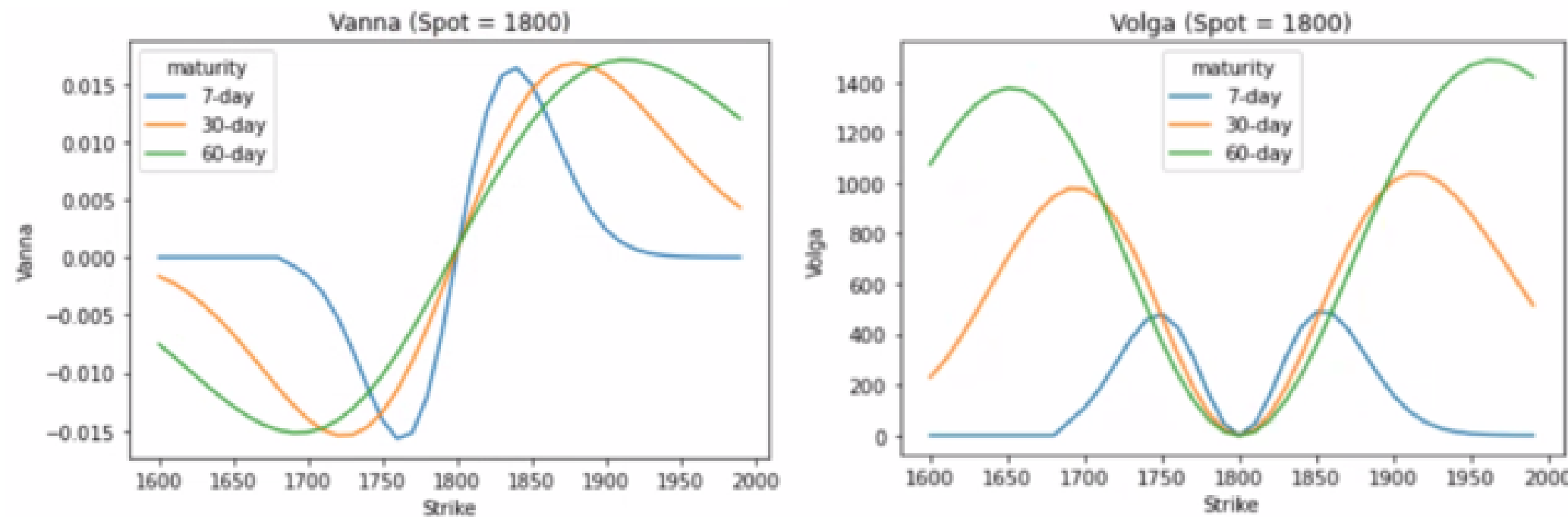
- **Vega:** Measure option price's sensitivity to implied volatility



- Vega is always **positive**

Delta Prime and Vega Prime

- **Delta Prime (Vanna):** Measure option **delta's** sensitivity to implied volatility
- **Vega Prime (Volga):** Measure option **vega's** sensitivity to implied volatility



- Delta Prime is **positive for Call Strike** and **negative for Put Strike**
- Vega Prime is always **positive**

Bonus Question

Gold trading at 1800 and 1900 put has a 90 delta (-0.9), how does the delta change if the implied vol increases?

Psi

Definition

Psi measures option price's sensitivity to the slope of the skew.

Measure Risk

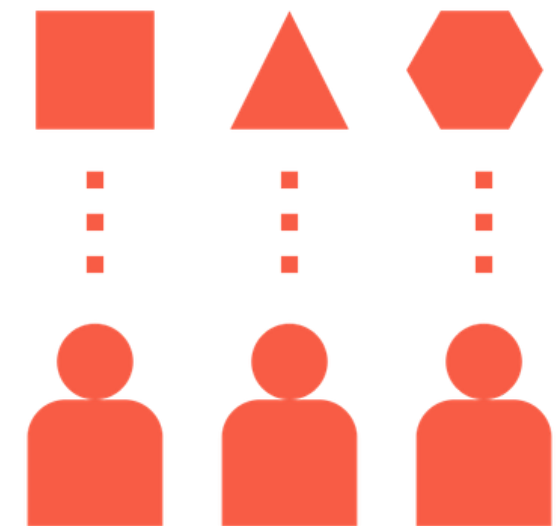
Each desk has **different ways to measure** this risk.

Call and Put

Call has positive Psi and **Put** has negative Psi.

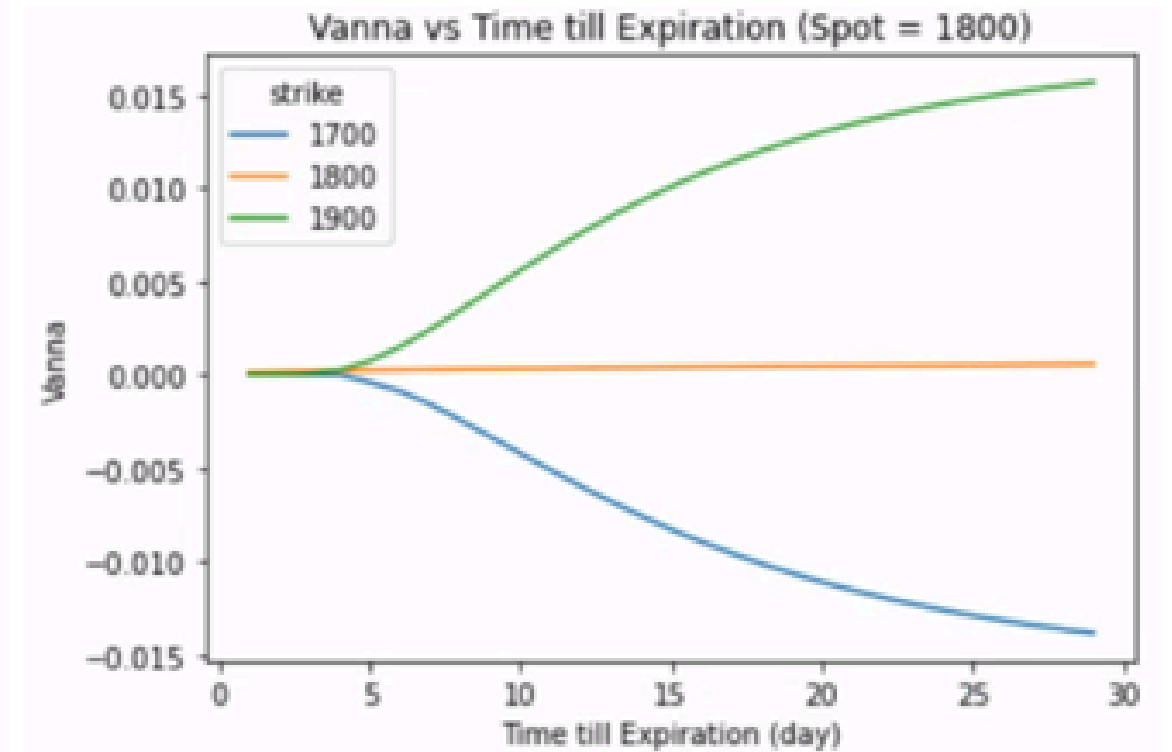
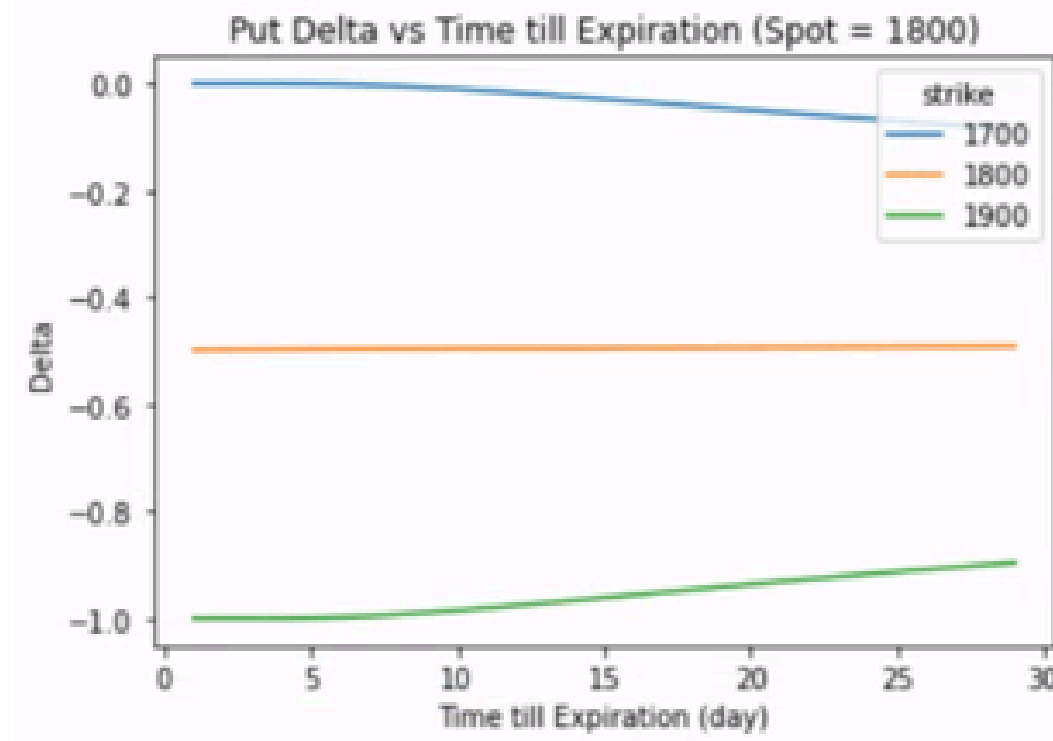
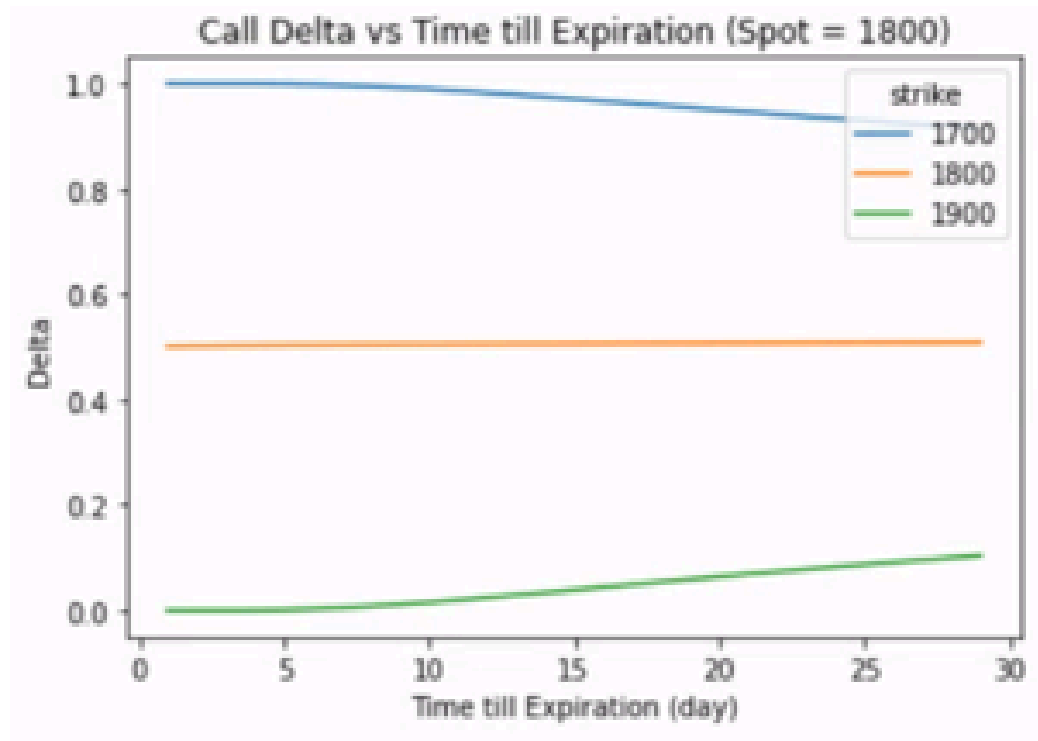
Characteristic of Greeks

- As an **option portfolio manager**, it's not only important to think about your current risk, but also how your risk changes in respect to the change in the underlying, the implied vol, and the time.
- Try to think about them **conceptually**, instead of forcing yourself to memorize the relationship.
- **Don't be frustrated!** It could take option traders a long time to master the concept. We are doing it in two hours!



Greeks vs Time

- Delta vs Time, Delta Prime vs Time



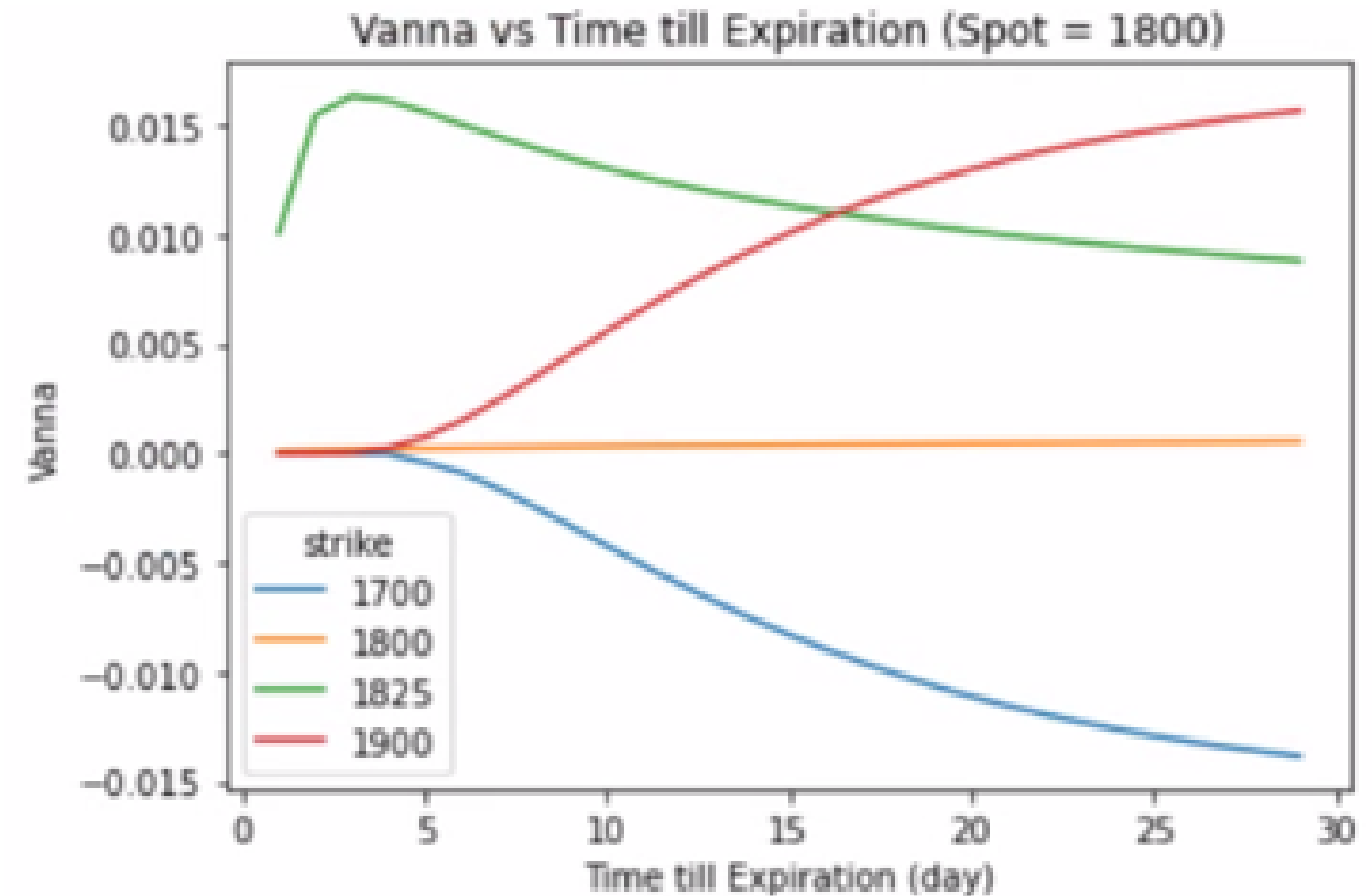
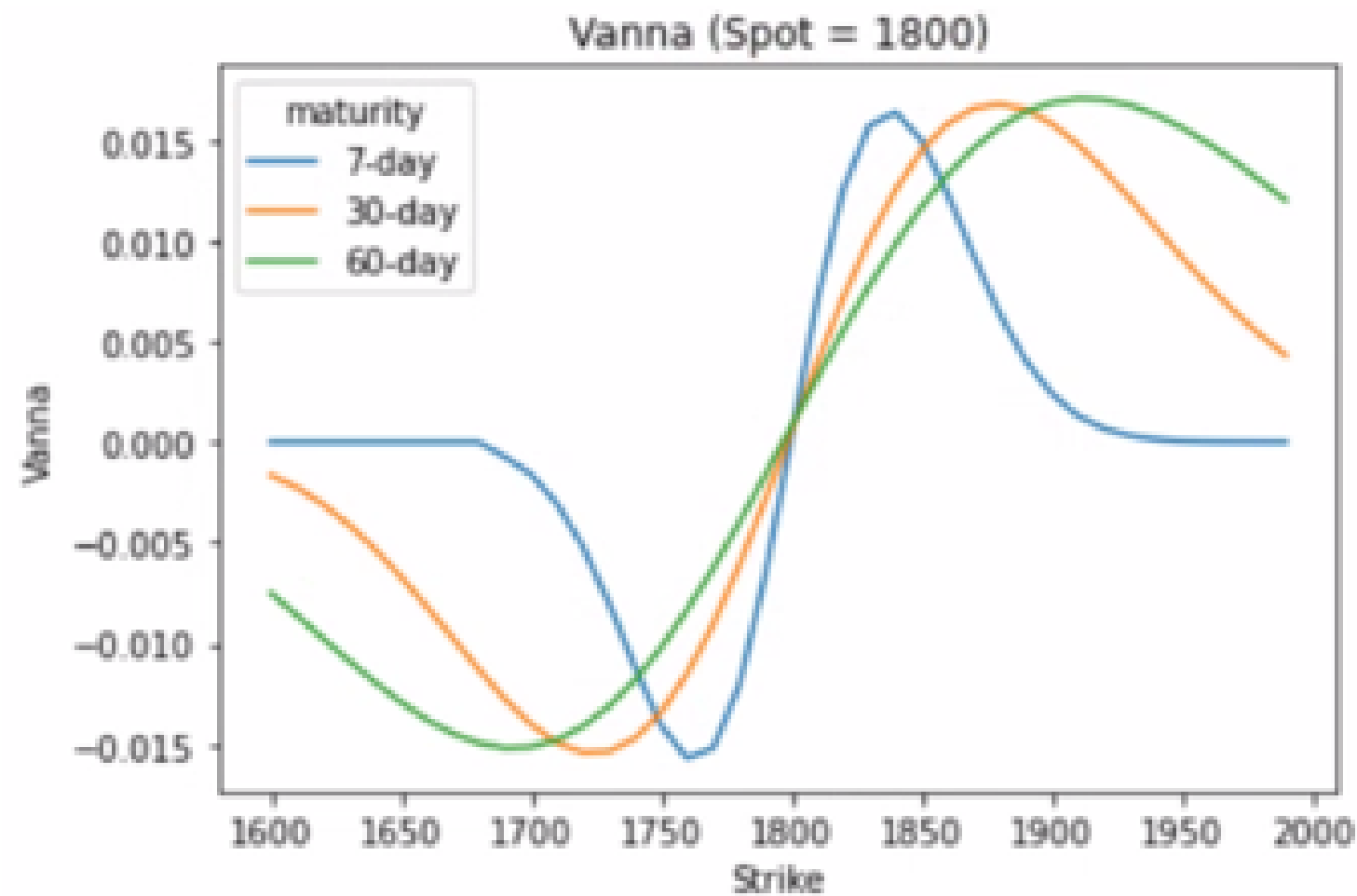
Bonus Question

Gold is trading at 1800 and the 1825 call of a 30-day option has a 45 delta (0.45). Assuming the spot stays the same, how does the **delta prime** of the 1825 call change as we approach expiration?

Greeks vs Time

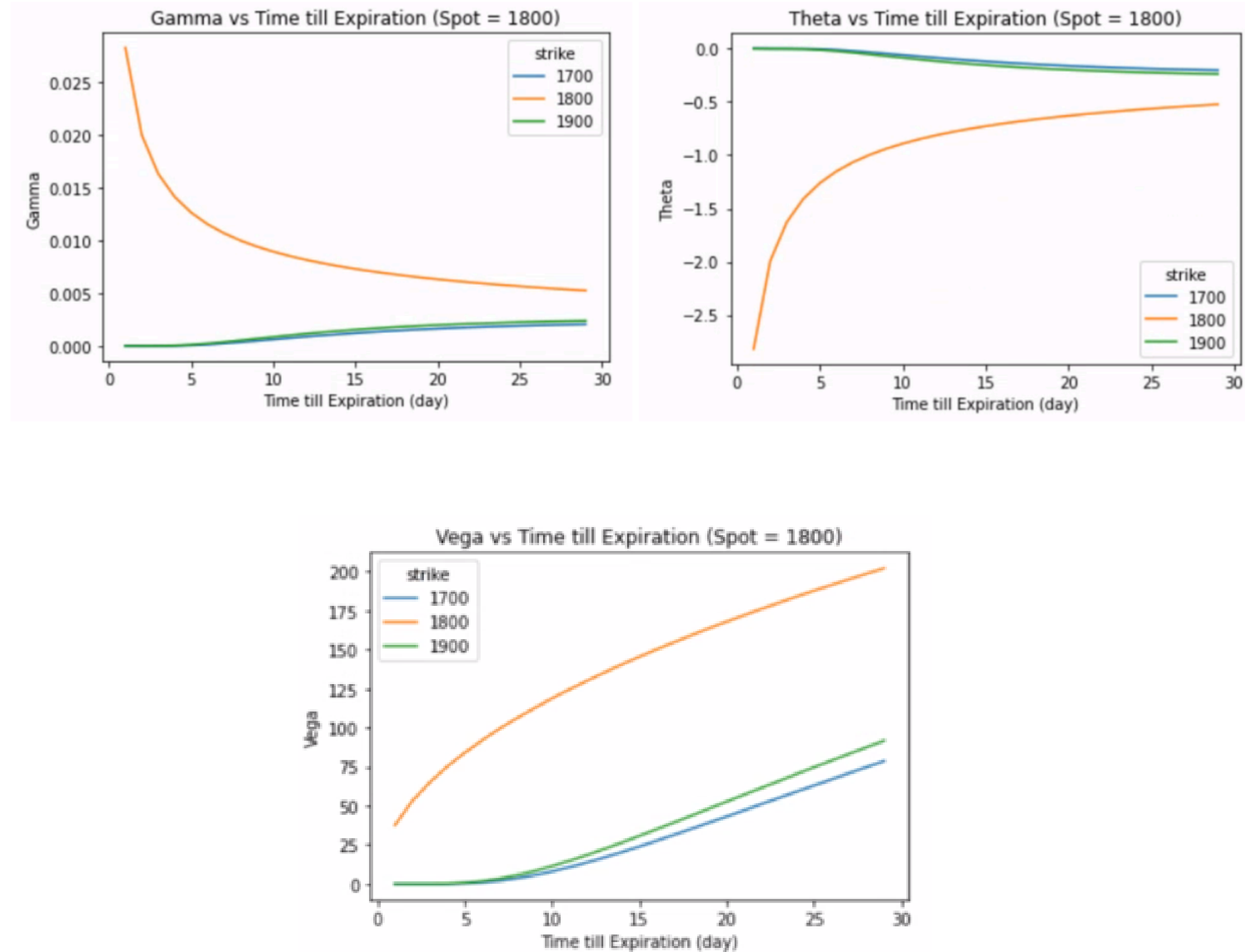
Answer to Bonus Question!

- It first goes up, then reaching a peak, and eventually goes down to zero.



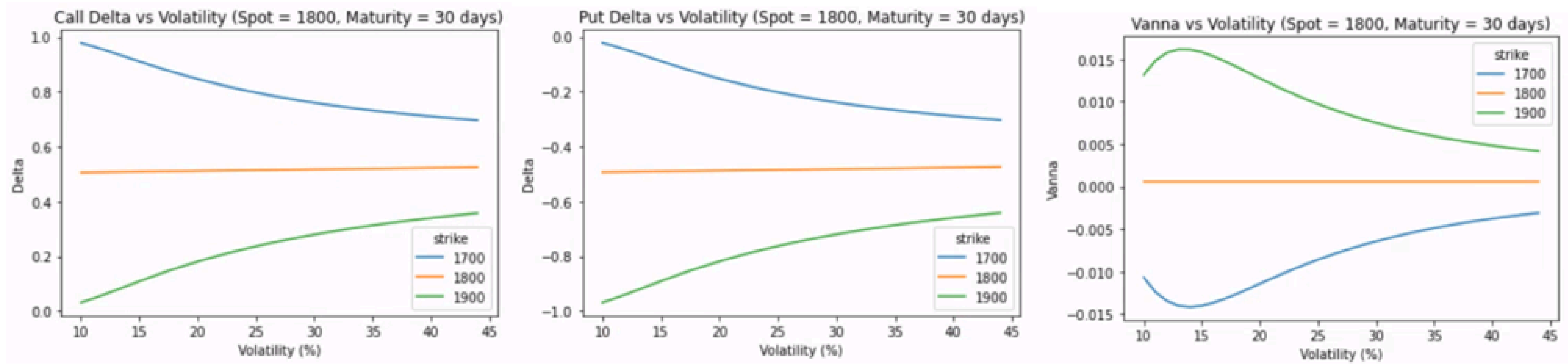
- Vega Prime (Volga) has similar behavior.

Greeks vs Time



Greeks vs Implied Volatility

Delta vs Vol, Delta Prime vs Vol

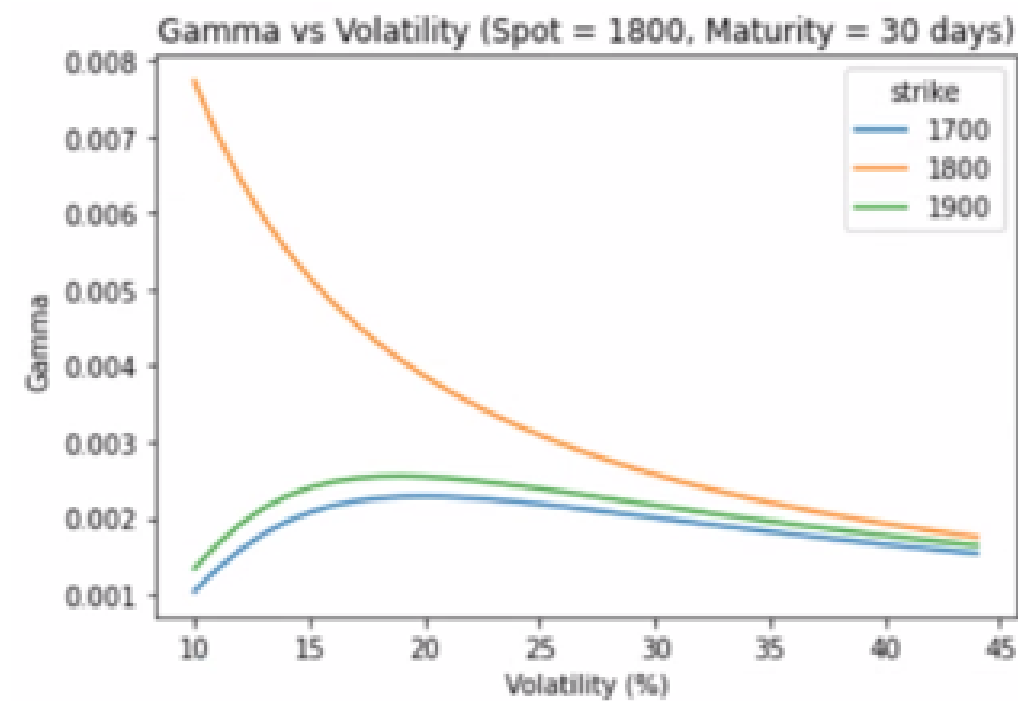


Bonus Question

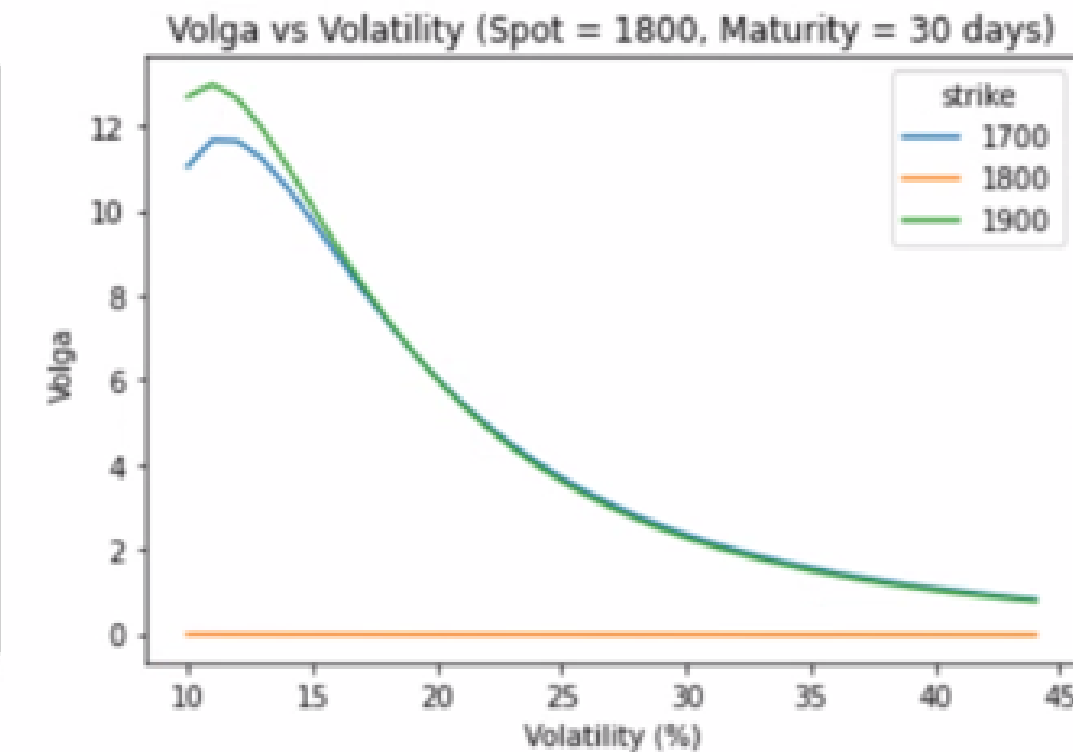
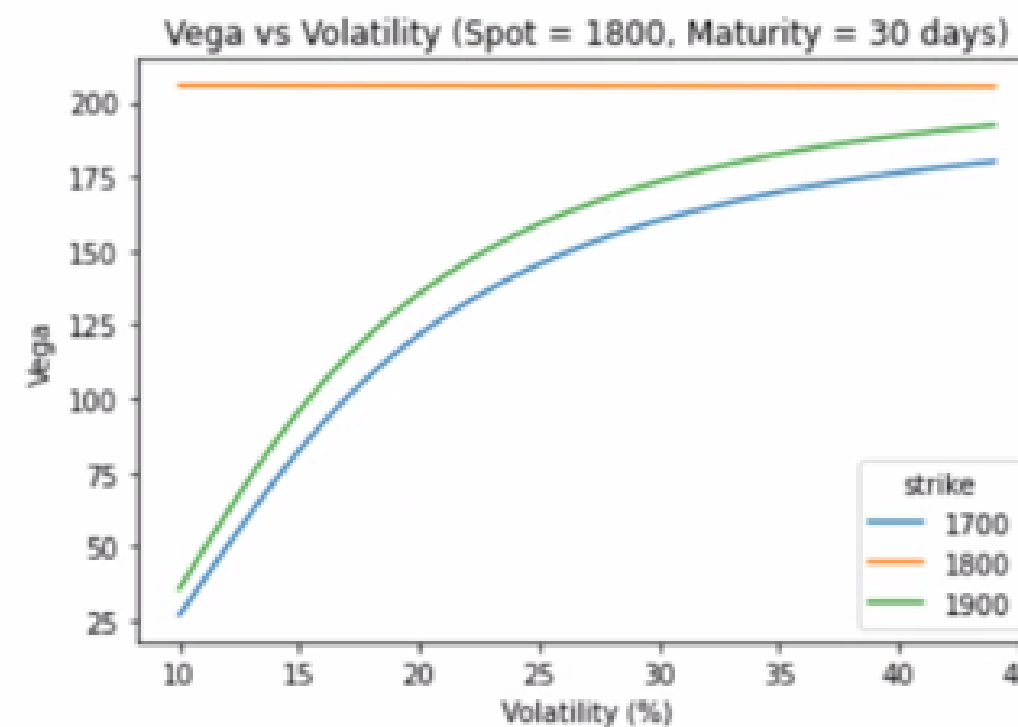
Can you explain conceptually the reason behind the shape of Vanna vs Vol Chart?

Greeks vs Implied Volatility

Gamma vs Vol, Theta vs Vol



Vega vs Vol, Vega Prime vs Vol



1. You own a one-month 1000 strike call and the delta is hedged. The current underlying is trading 1000 and the implied vol is 15%.

a. What are the greeks of your current portfolio? (positive, negative, roughly zero, hard-to-tell)

Delta	Gamma	Theta	Vega	Delta Prime	Vega Prime

b. The underlying has rallied to 1050, what are the greeks of your portfolio now? (positive, negative, roughly zero)

Delta	Gamma	Theta	Vega	Delta Prime	Vega Prime

c. How would the greeks have changed when the spot rallied from 1000 to 1100? (increasing, decreasing, roughly unchanged, hard-to-tell)

Delta	Gamma	Theta	Vega	Delta Prime	Vega Prime

2. What would most likely happen to the greeks of the one-month 1050 call, assuming the underlying is trading at 1000 and the call has a 25 delta (0.25)? (increasing, decreasing, roughly unchanged, hard-to-tell)

	Delta	Gamma	Theta	Vega	Delta Prime	Vega Prime
Implied Vol Increases						
Five days have gone by						

Topic 4: In-Class Exercise

Topic 5: Common Option Spreads

Option Spreads

Trading **Option Spread** involves buying and selling more than just a single option.

Question: What do we need to think about when we (**the market maker**) trade an option spread?



Option Spread PnL at Expiration

Description

A common option spread is Call Spread. It involves buying a call and selling another call with higher strike at the same expiration day.

Example

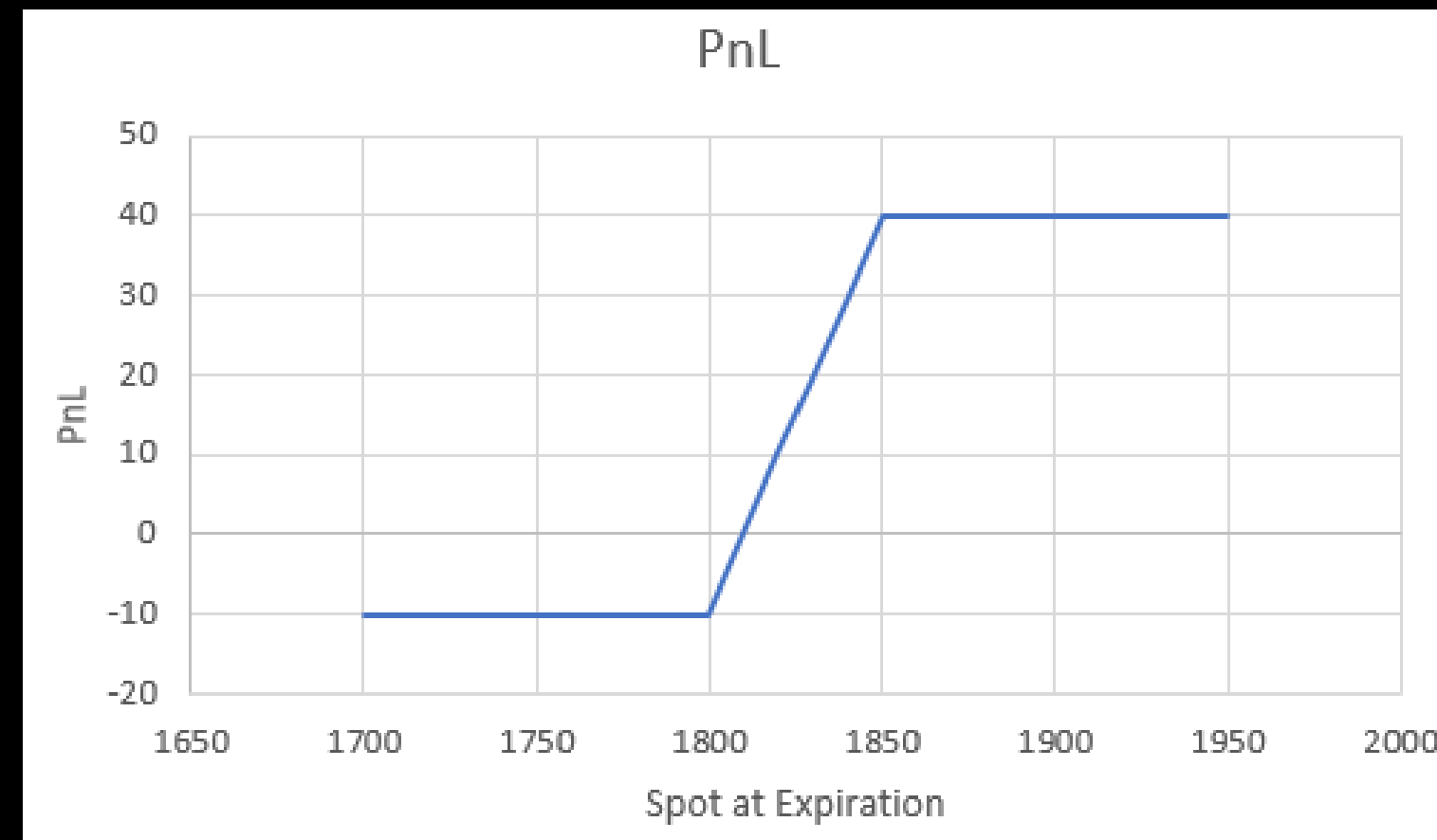
For example, you pay \$20 for one 1800 call expiring in one month and sell one 1850 call at \$10.

Questions

- What's the maximum profit and loss?
- What does the PnL chart look like at the expiration? (Y is PnL, X is Spot)

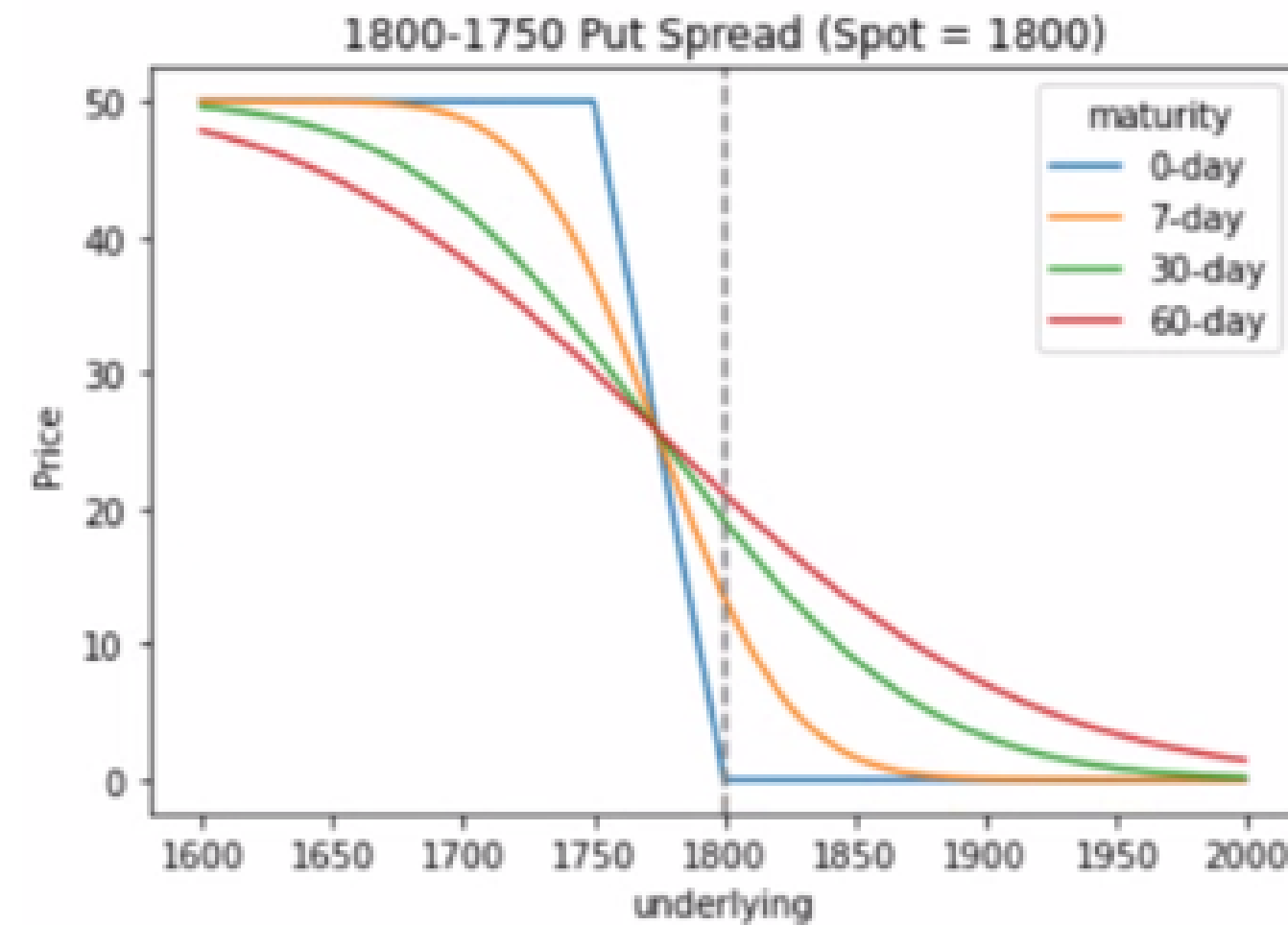
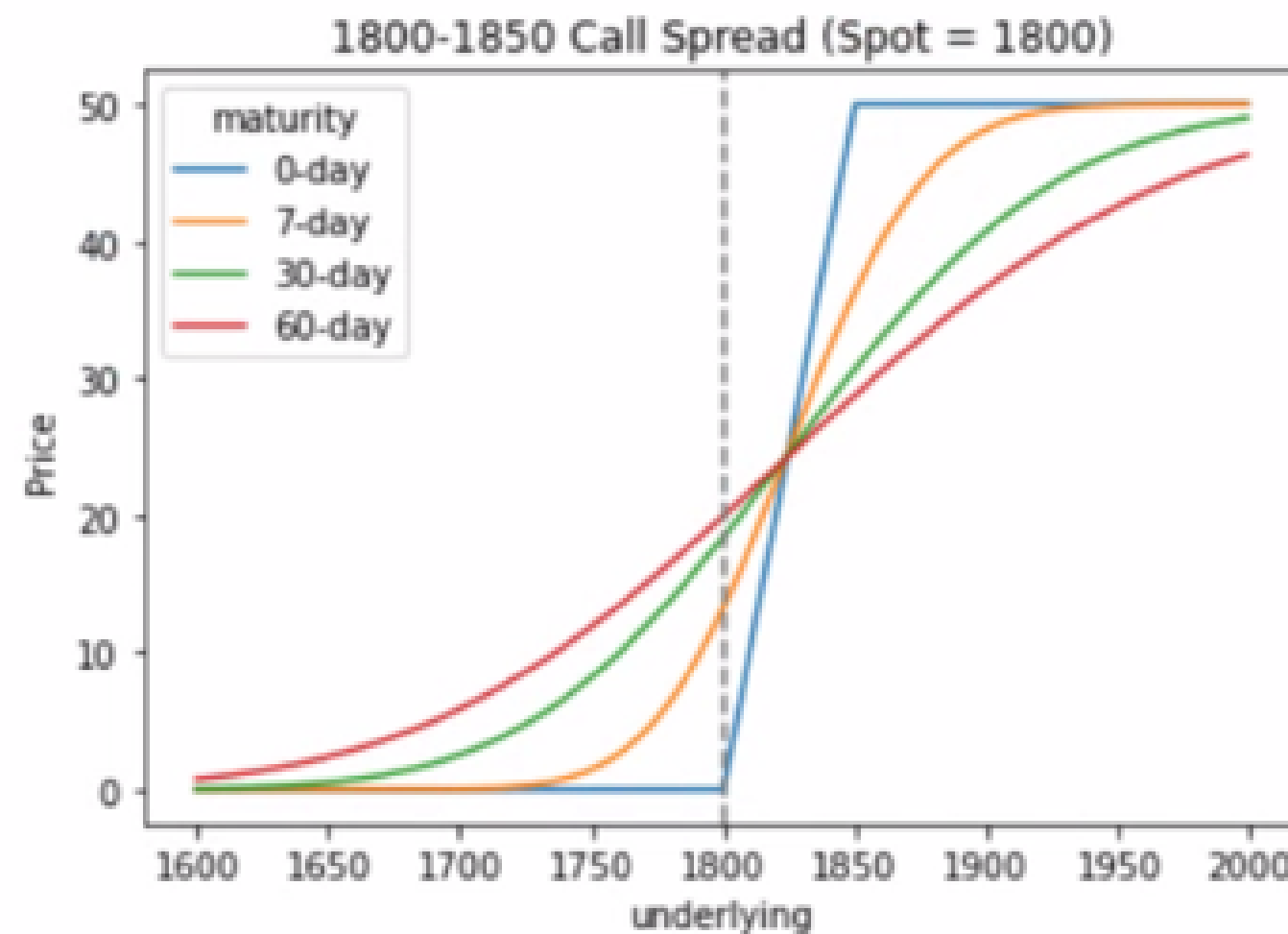
Answer

- **Max Loss** is \$10 (the premium you paid).
- **Max Profit** is \$40.
- **Breakeven Spot** is 1810.



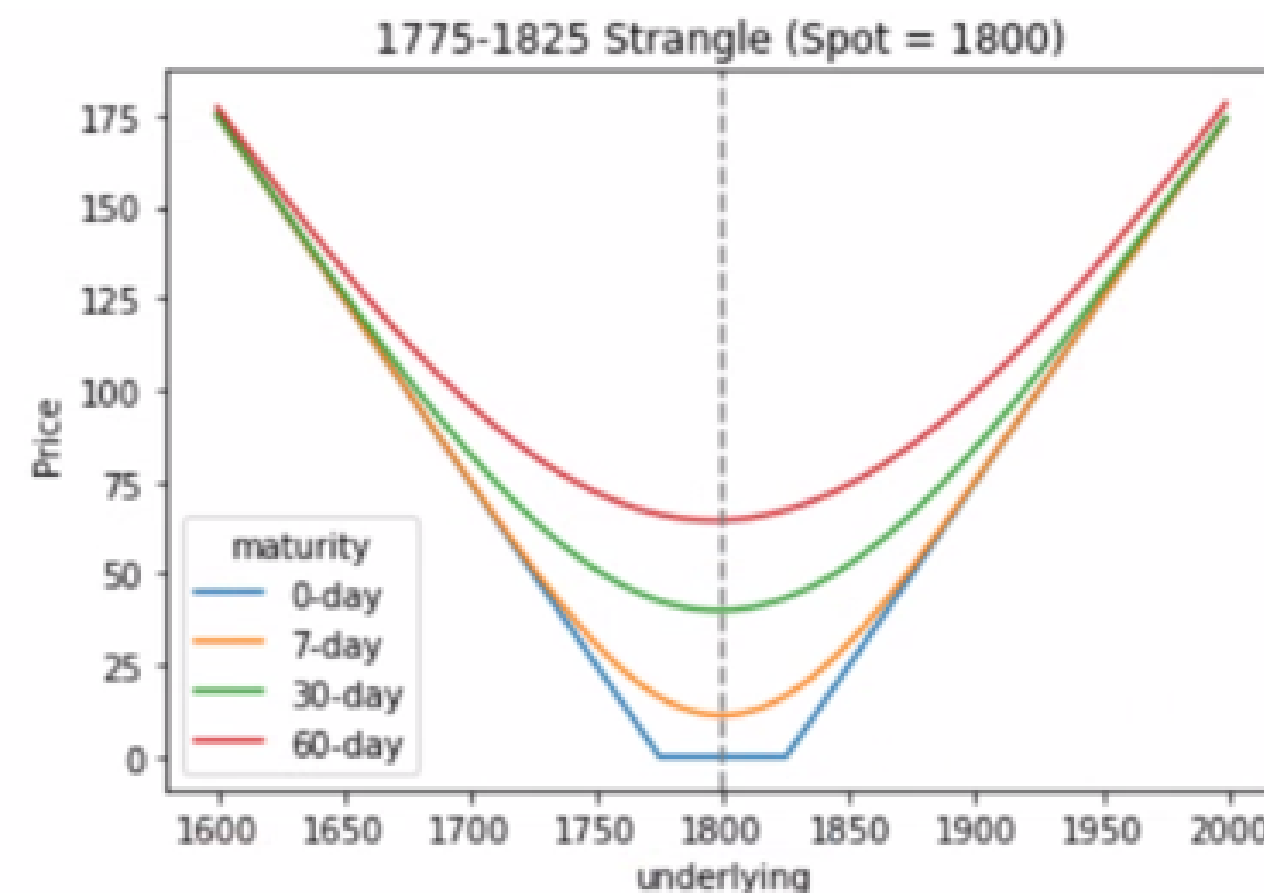
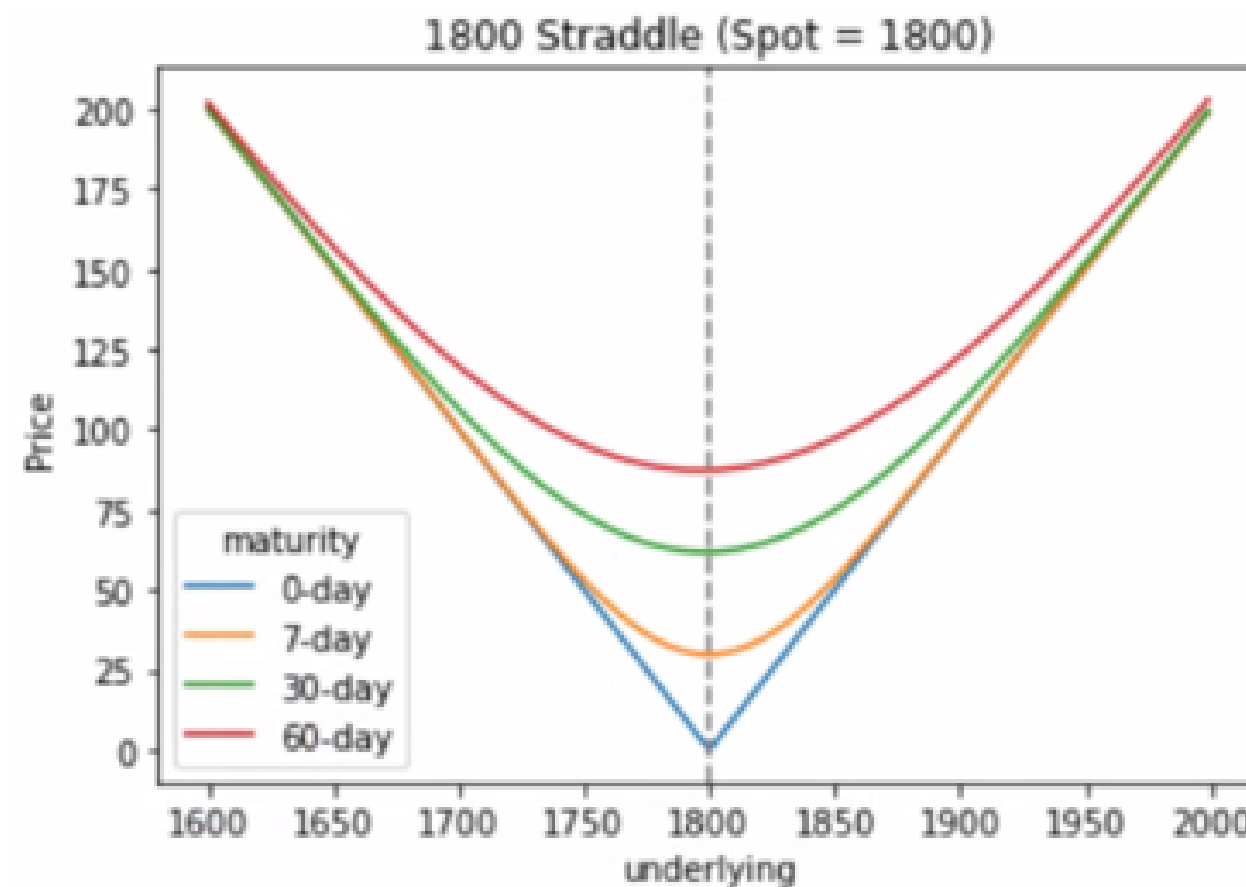
Call Spread and Put Spread

- **Call Spread:** Buy 1 call and Sell 1 call with a higher strike
+1 1800 Call, -1 1850 Call
- **Put Spread:** Buy 1 put and Sell 1 put with a lower strike
+1 1800 Put, -1 1750 Put



Straddle and Strangle

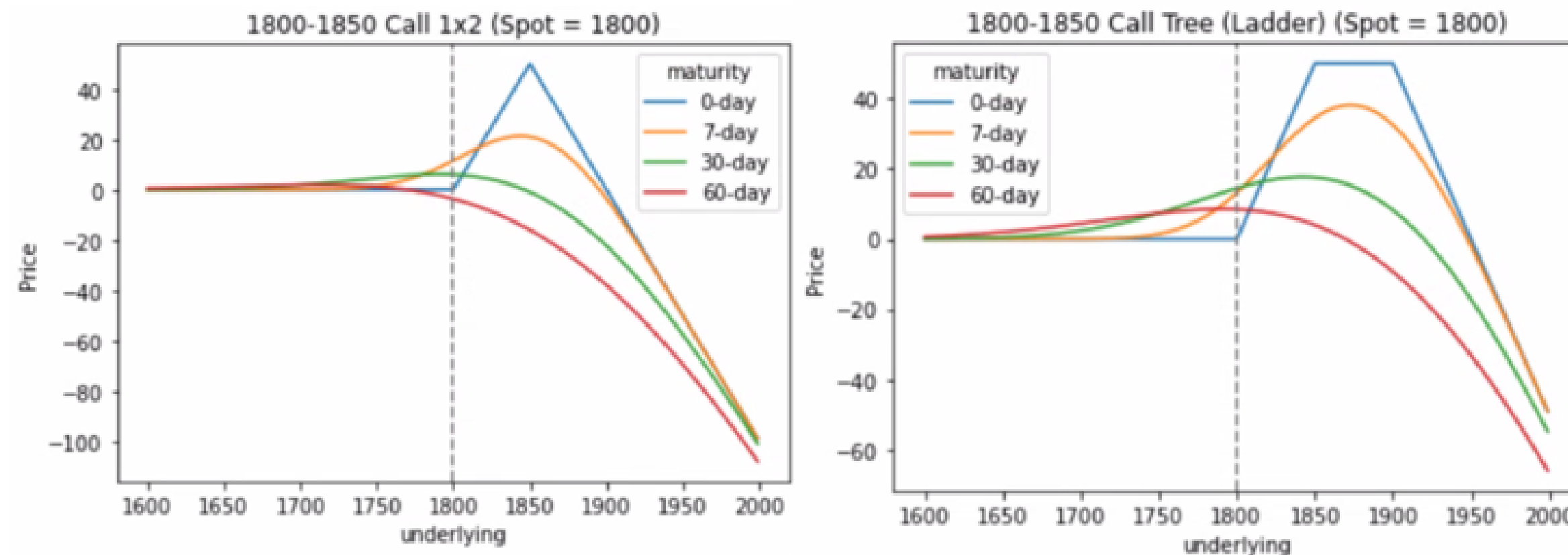
- **Straddle:** Buy 1 call and 1 put with SAME strike
+1 1800 Call, +1 1800 Put
- **Strangle:** Buy 1 call and 1 put with DIFFERENT strike
+1 1825 Call, +1 1775 Put



- **Bonus Question:** Assuming both straddle and strangle expire at the same time, which spread should cost you more to buy?

Call Spread 1x2 and Call Tree/Ladder

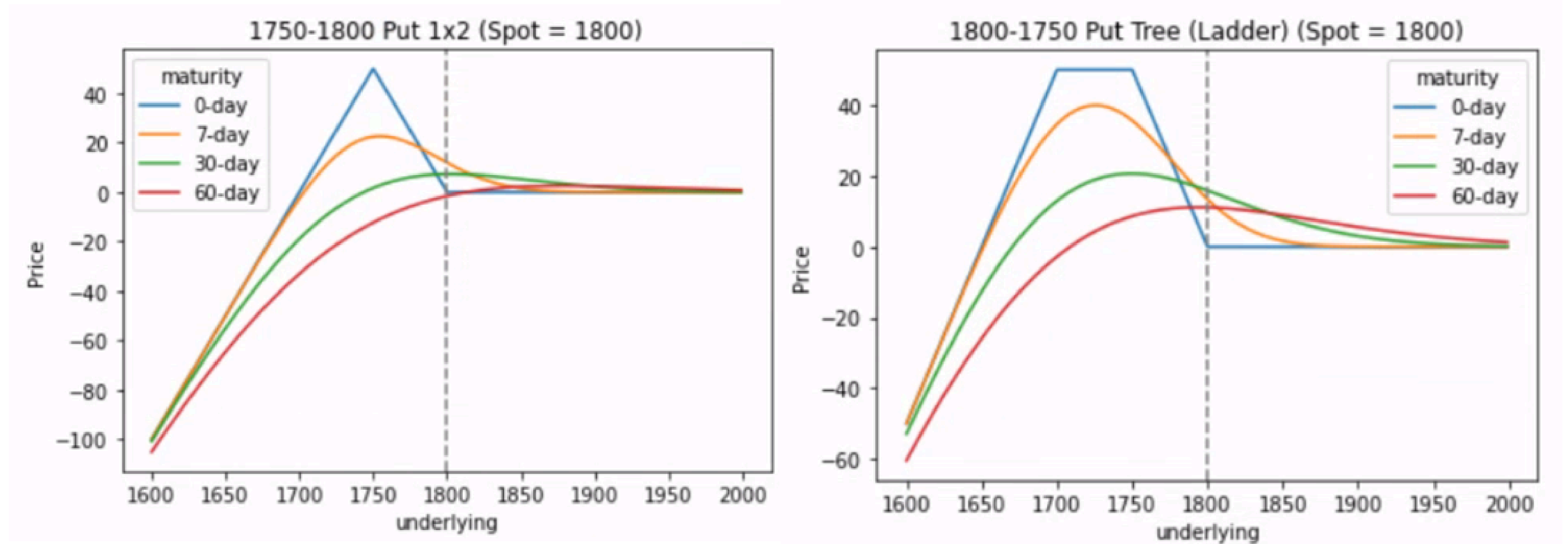
- **Call Spread 1x2:** Buy 1 call and Sell 2 call with a higher strike
+1 1800 Call, -2 1850 Call
- **Call Tree:** Buy 1 call, Sell 1 call with a higher strike, and Sell 1 more call at an even higher strike
+1 1800 Call, -1 1850 Call, -1 1900 Call



- **Bonus Question:** Can you draw the 1800-1750 Put Spread 1x2 and 1800-1750-1700 Put Tree PnL chart? At the expiration, at which spot do you earn the max profit? What's your max loss? Where is your breakeven (assuming you buy the spread at a price of zero)?

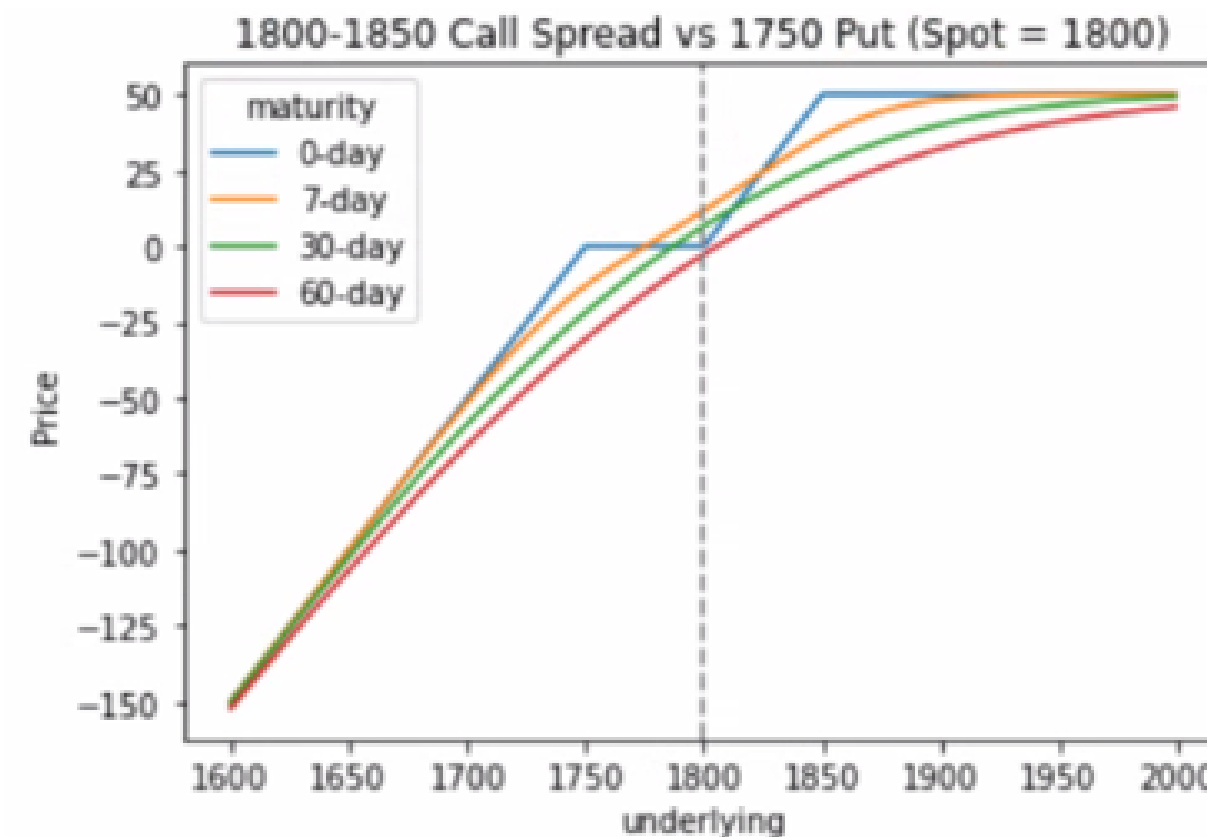
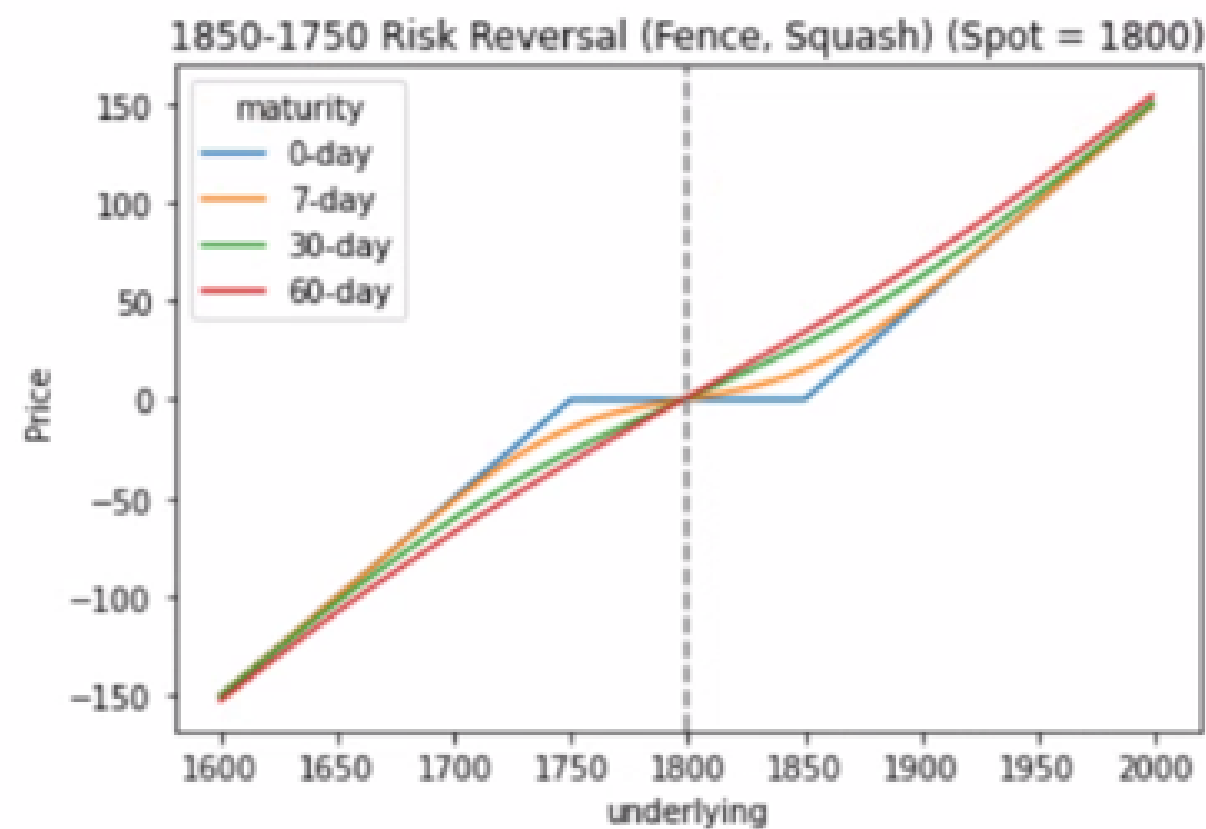
Call Spread 1x2 and Call Tree/Ladder

Answer to Bonus Question!



Risk-Reversal and Call-Spread vs Put

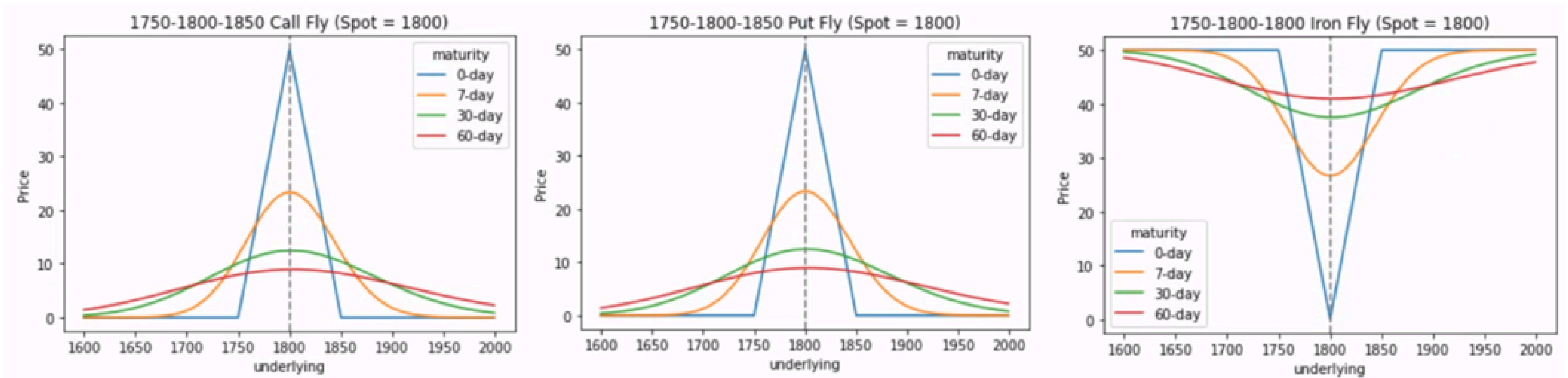
- **Risk Reversal:** Buy 1 call and Sell 1 put
+1 1850 Call, -1 1750 Put
- **Call Spread vs Put:** Buy 1 call, Sell 1 call with a higher strike, and Sell 1 put with lower strike
+1 1800 Call, -1 1850 Call, -1 1750 Put



- **Put-Spread vs Call** is another variation of the Call-Spread vs Put. Just like the 1x2 and the Tree, you are selling additional option of your choice to reduce the premium of your spread.

Call Fly, Put Fly, and Iron Fly

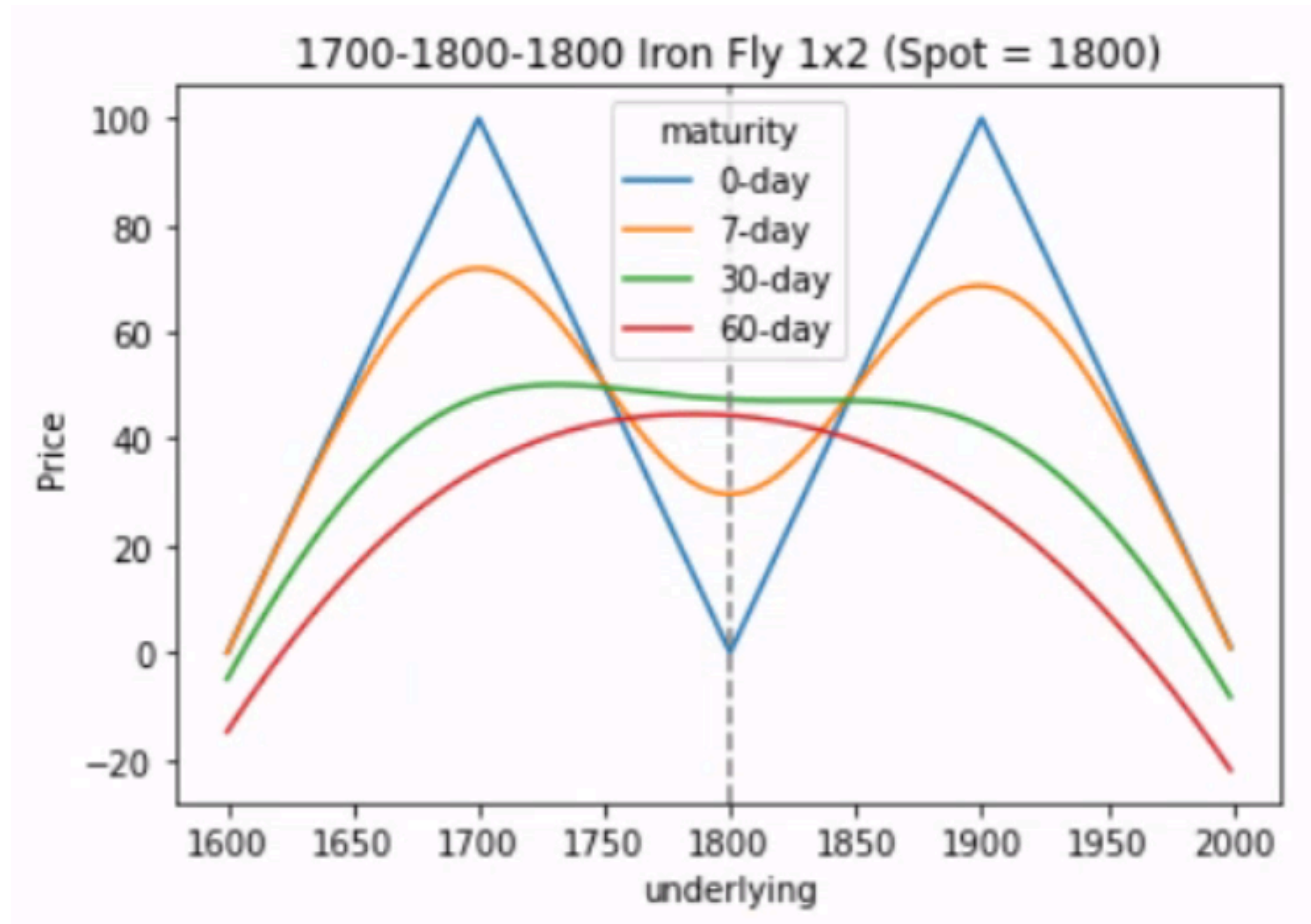
- **Call Fly:** Buy 1 call, Sell 2 calls of higher strike, and Buy 1 call at even higher strike
+1 1750 Call, -2 1800 Call, +1 1850 Call
- **Iron Fly:** Buy 1 straddle of middle strike and Sell 1 strangle of the outside strikes
+1 1800 Straddle, -1 1750/1850 Strangle



- **Bonus Question:** If you actually pay \$20 for the Call Fly above, what's your max profit and loss? Where in spot(s) is your breakeven point? What price must the Iron Fly be trading at to avoid arbitrage opportunity?

Ratioed Iron Fly

- **Iron Fly 1x2:** Buy 1 straddle of middle strike and Sell 2 strangle of the outside strikes
+1 1800 Straddle, -2 1700/1800 Strangle



Topic 6: Types of Option Risks

And, How to Trade Them Using Option Spreads

Choosing the Option Strategy

What option strategy would you likely choose if...

- Vol price is too cheap
- The market is likely to have an “unexpected” big move in the upcoming days
- Spot is range bound
- There is a very high chance spot “could” have a big rally due to geographic uncertainty, but vol is priced too expensive
- Spot is going to slowly drift lower over the next one-month
- There is a VERY small chance that spot is going to \$10,000 even though it’s only trading \$1800



Topic 7: Greeks Derivation from Option Spreads

Greeks Derivation

Market Maker

As a **Market Maker**, we are often delta-hedged and focus on trading the remaining portfolio risk.

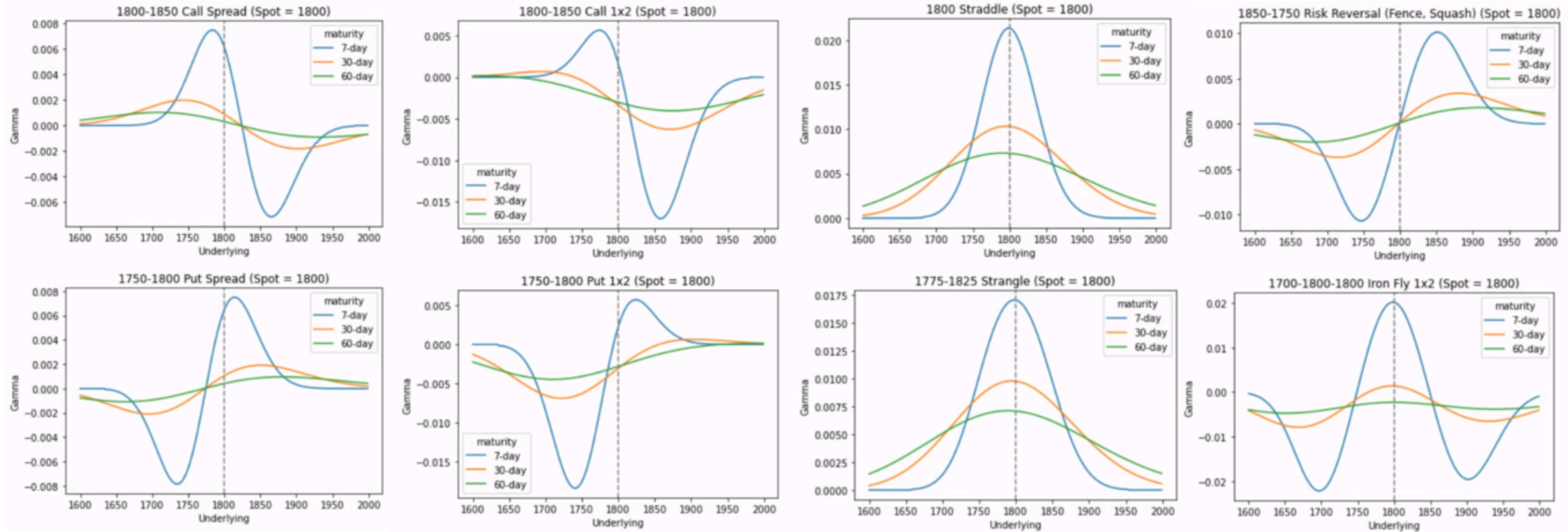
Payoff of a Delta-Hedged Option Spread

The **payoff of a delta-hedged option spread** can be significantly different than the payoff of the same option spread without delta-hedging.

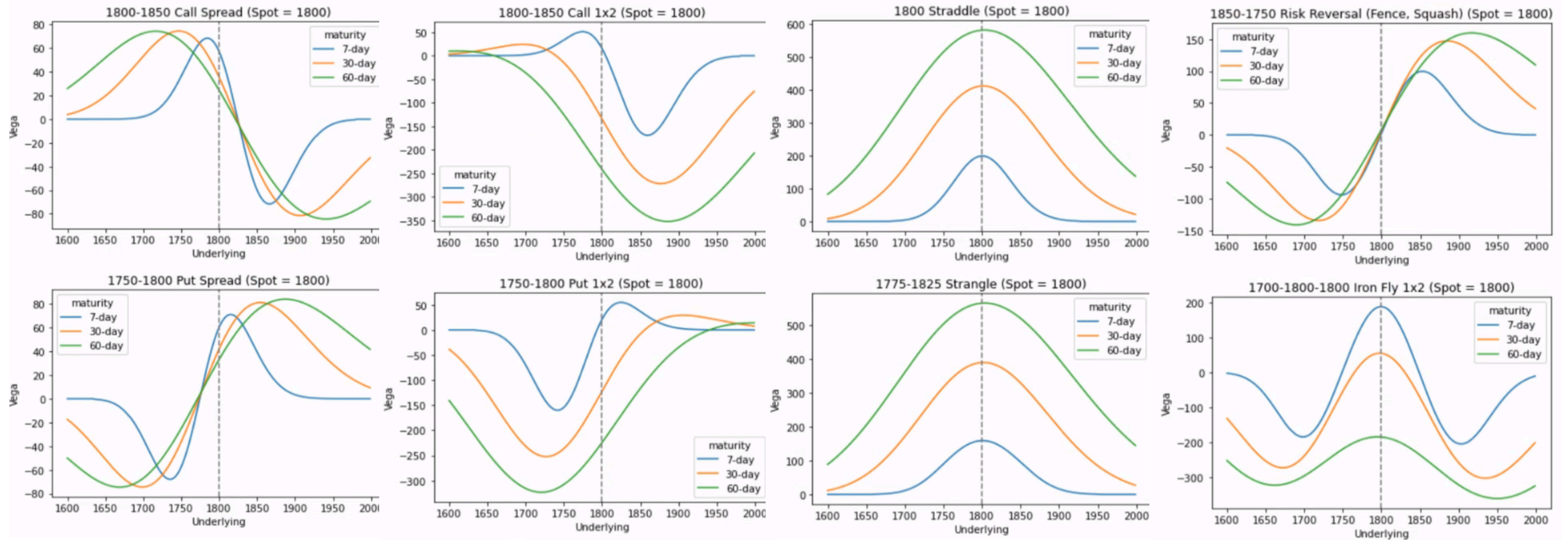
Look Beyond Price Changes

We also need to **look beyond just the price changes** and constantly think about how Greeks change in our portfolio.

Gamma Derivation Examples

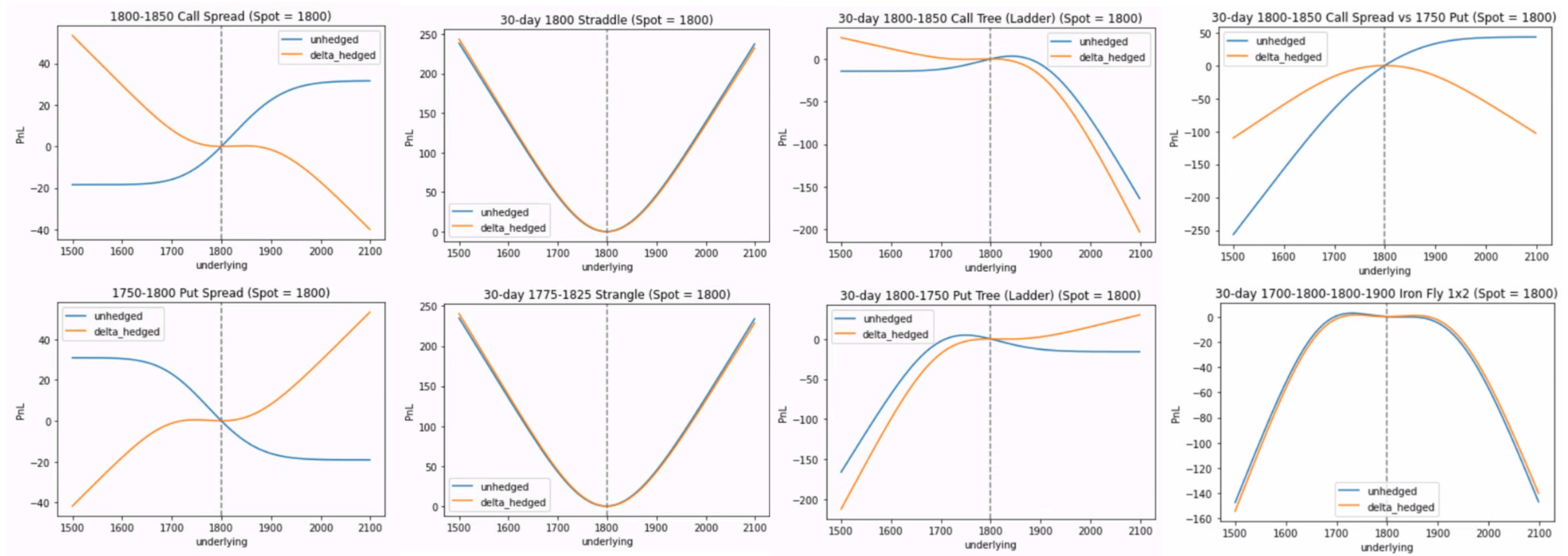


Vega Derivation Examples



Topic 8: PnL of Delta-Hedged Option Spreads

PnL of “Delta-Hedged” Option Spread



1. You own a one-month 1000 – 1050 call spread and the delta is hedged. The current underlying is trading 1000.

a. What are the greeks of your current portfolio? (positive, negative, roughly zero, hard-to-tell)

Delta	Gamma	Theta	Vega	Delta Prime	Vega Prime

b. The underlying has rallied to 1025, what are the greeks of your portfolio now? (positive, negative, close to zero)



Delta	Gamma	Theta	Vega	Delta Prime	Vega Prime

c. How would the greeks have changed when the spot rallied from 1000 to 1025? (increasing, decreasing, close to unchanged, hard-to-tell)

Delta	Gamma	Theta	Vega	Delta Prime	Vega Prime

d. The underlying continues and rallies to 1050, what are the greeks of your portfolio now? (positive, negative, close to zero)

Delta	Gamma	Theta	Vega	Delta Prime	Vega Prime

e. How would the greeks have changed when the spot rallied from 1000 to 1050? (increasing, decreasing, close to unchanged, hard-to-tell)

Delta	Gamma	Theta	Vega	Delta Prime	Vega Prime

THE END